Shepherd

Real-Time Adaptive Wildfire Evacuation

Meet the Team



Harsit

Electrical Engineering



Azer

Accounting



DV

Statistics



Shriyaz

Computer Science



Aarya

Computer Science

“When the city runs, Shepherd leads.”

How can we make wildfire evacuations safer, smarter, and more accessible?

Which roads will be accessible?

Which shelters will be safe?

When should you leave?

How can we minimize gridlock? (this is key)

The Gap

Current State

- Emergency alerts
- Separate information sources
- Changing road conditions

Filling the Gap

Desired State

- Unified platform
- Dynamic rerouting
- Safer evacuation planning

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Adaptive evacuation routing

Desired State

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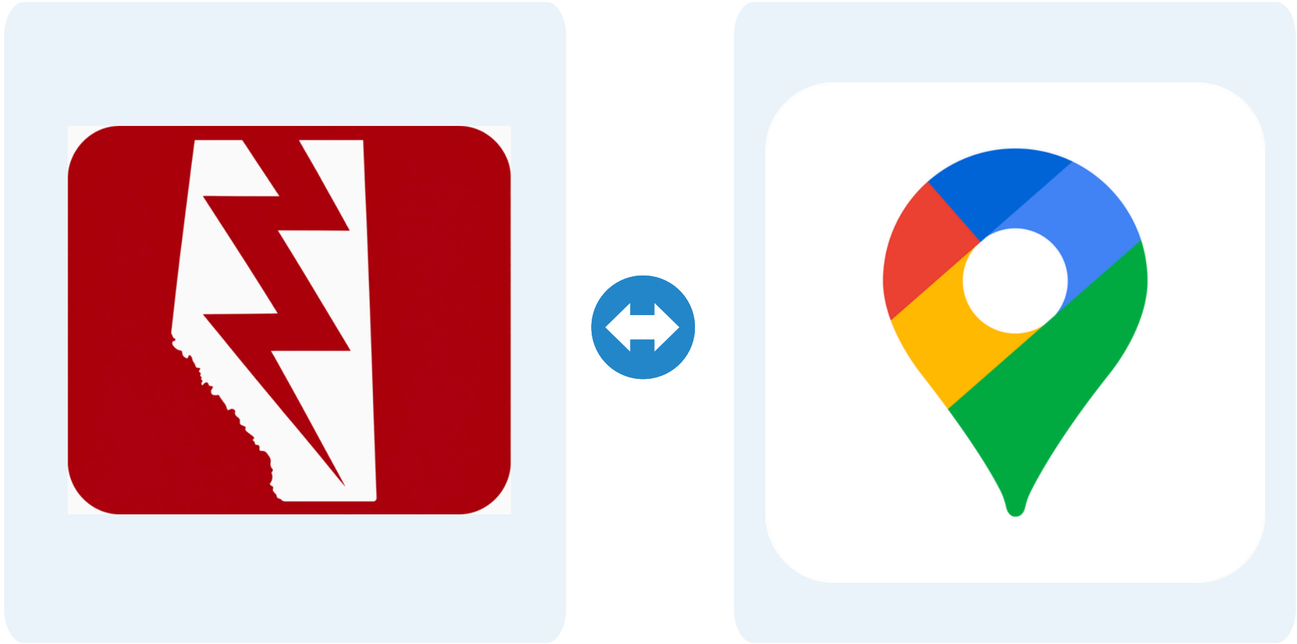
Shepherd addresses the changing conditions between the warning and the next move.

What does Alberta Currently Have?

Emergency Alerts

Navigation Apps

What if they work together?



One platform for alerts, routing, and response.



Shepherd

Here’s how we’ll do it.

Live Demo

What impact will Shepherd have?

Why Shepherd Matters.

For Residents

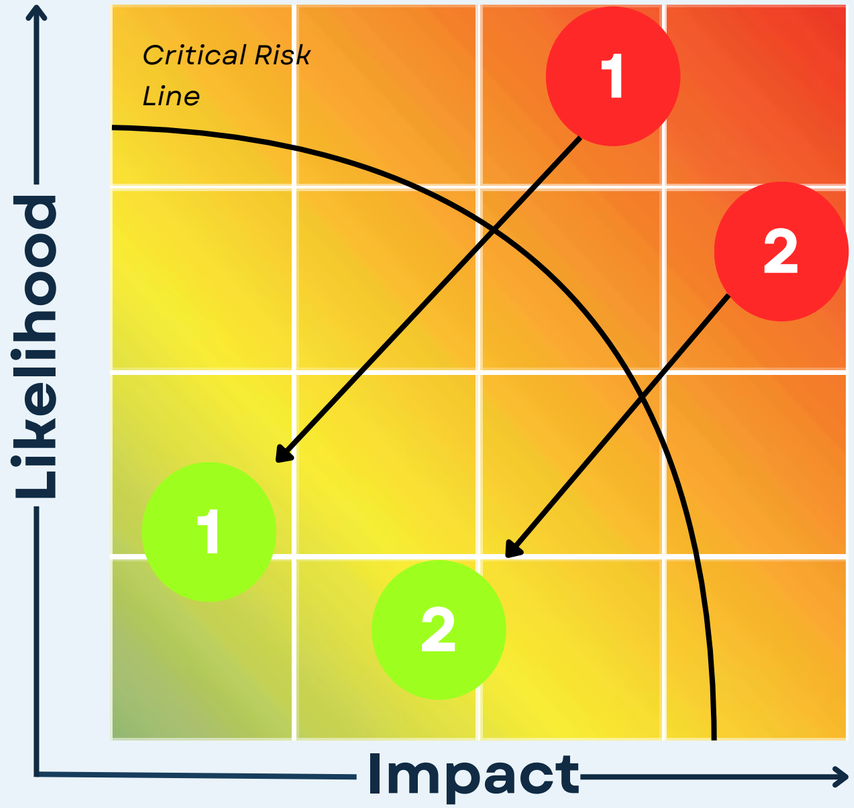
- Adaptive evacuation routes
- Accessible shelter information
- More informed decisions

For Municipalities

- Emergency scenario testing
- Identify potential traffic congestion
- Improved evacuation coordination

Risks and Mitigations

Risk Matrix



Risk	Mitigation
Inaccurate hazard predictions	Verified emergency data + expert validation
System or network failures	Reliability testing + offline capabilities

PROTOTYPE TO DEPLOYMENT

Prototype

1

Baseline simulation and road network

Validation

2

Verified data, safety testing and agency feedback, initial funding

Municipal Pilot

3

Partner with municipalities to test and refine (B2G2C)

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